MEDALLION AZURE ADF DBT DATABRICKS

Modern Data Engineering with Azure Databricks + dbt + Unity Catalog + ADF

1. PROJECT OVERVIEW

This project implements a **production-grade Medallion Architecture** (Bronze → Silver → Gold) on Azure. It demonstrates the complete modern data platform lifecycle from raw source data to governed, business-ready analytical models.

Objective: Build a scalable, governed Lakehouse that ingests data from Azure SQL, processes it through Bronze/Silver/Gold layers using Unity Catalog for governance, dbt for transformations and snapshots, and Azure Data Factory for orchestration.

2. SYSTEM ARCHITECTURE

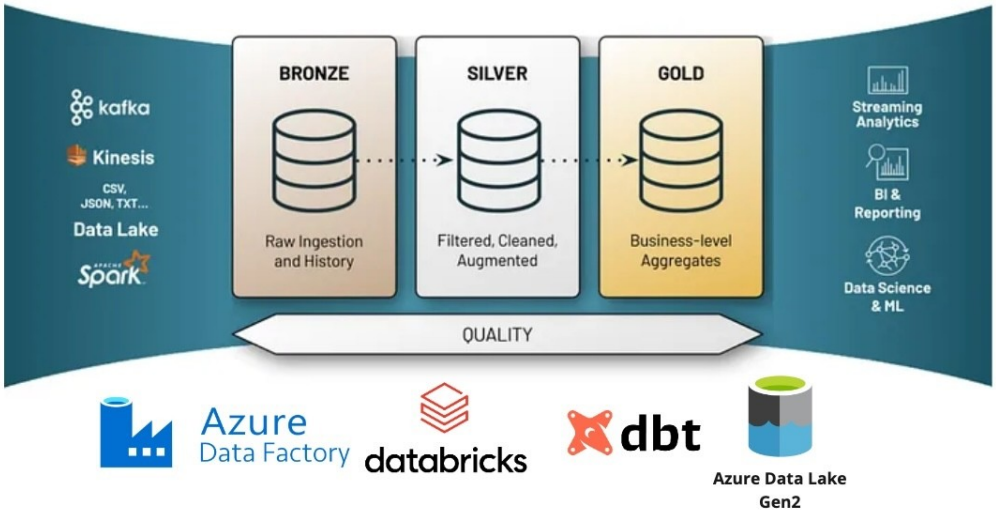

Figure 1: Medallion Architecture — Bronze (Raw) → Silver (SCD Snapshots) → Gold (Dimensional Models) with ADF, Databricks, dbt, and Unity Catalog

3. TECHNOLOGY STACK

Layer	Technology	Role
Source	Azure SQL Database (AdventureWorksLT)	Transactional source system
Orchestration	Azure Data Factory	Pipeline orchestration + Dynamic Notebook execution
Ingestion & Bronze	Azure Databricks + ADLS Gen2	Raw data landing + External tables in Unity Catalog
Silver (SCD)	dbt Snapshots	Slowly Changing Dimension Type 2 historical tracking
Gold (Modeling)	dbt Models (marts)	Dimensional modeling (dimensions + facts)

Governance	Unity Catalog	Centralized catalog, schemas, access control, lineage
Storage	Azure Data Lake Storage Gen2	Bronze, Silver, Gold containers + Unity Catalog metastore
Development	Databricks Notebooks + VS Code + dbt	Interactive development and debugging

4. PRE-REQUISITES — Azure Databricks Unity Catalog & Governance Setup

Before running any data pipeline, the following Unity Catalog infrastructure must be established:

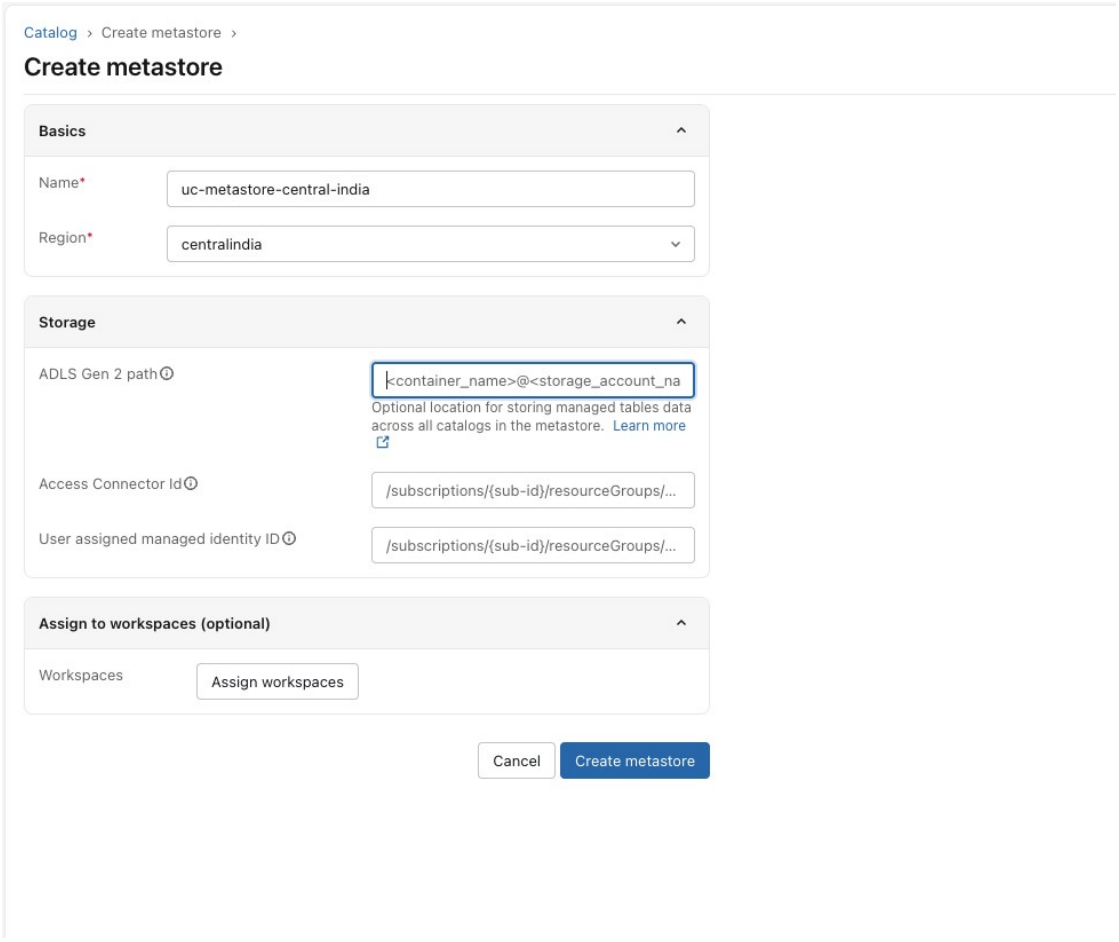

Figure 2: Unity Catalog Metastore created in Central India region

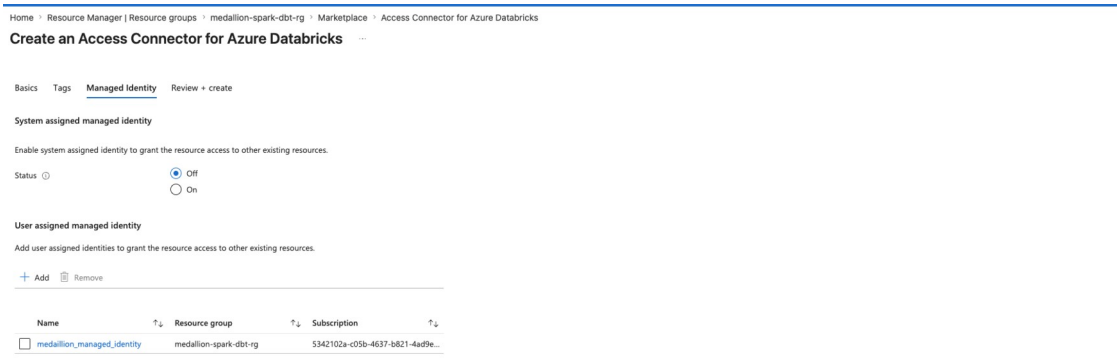

Figure 3: Access Connector for Azure Databricks linked to Managed Identity

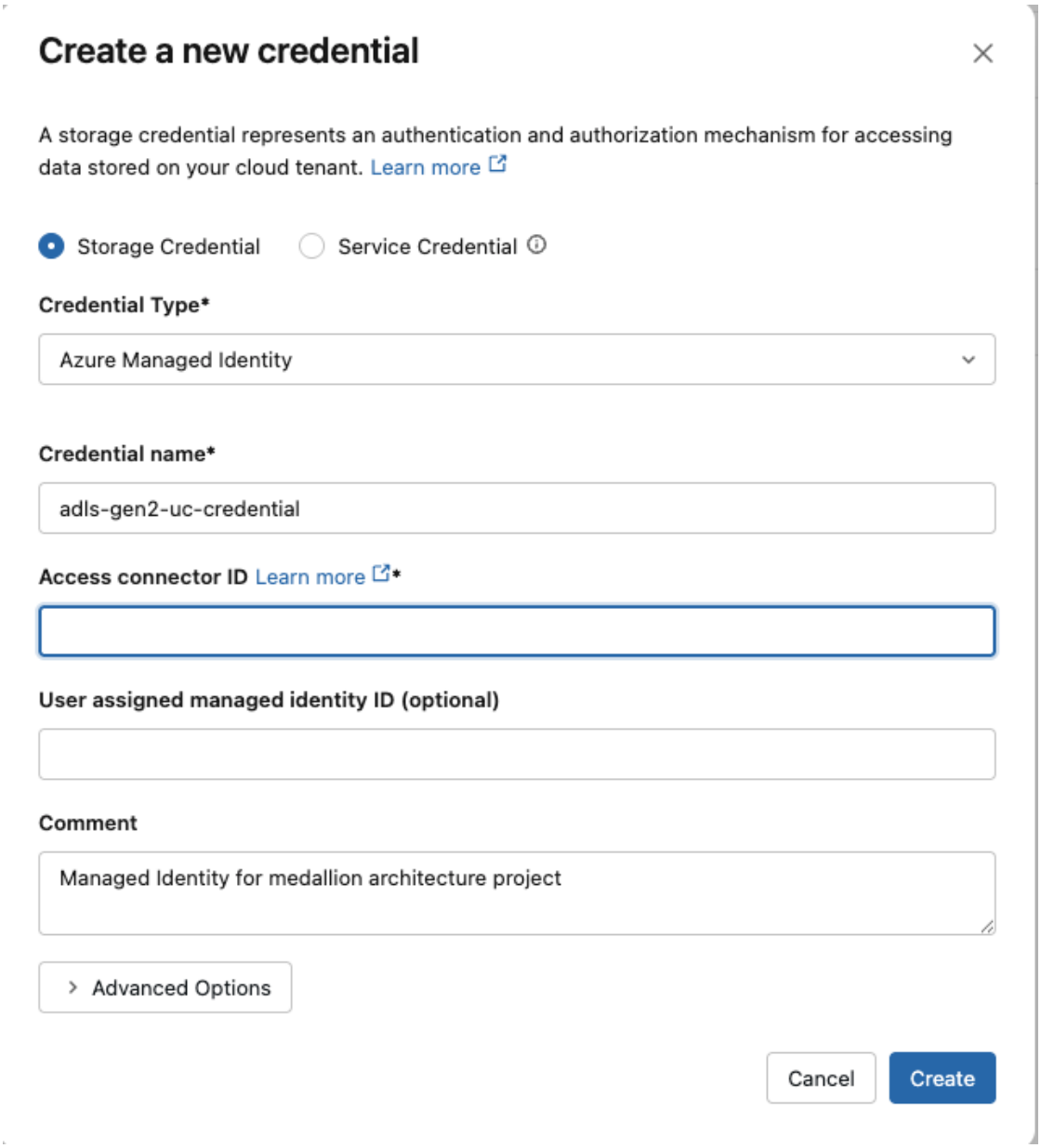

Figure 4: Storage Credential created for ADLS Gen2 access for external locations

- **Key Setup Steps Performed**:**
  - create Azure Databricks resource
  - Created Unity Catalog Metastore in Central India (http://accounts.azuredatabricks.net/)
  - Created User-Assigned Managed Identity (medallion_managed_identity)
  - Created Access Connector for Azure Databricks
  - Created Storage Credential (adls-gen2-uc-credential) using Managed Identity
  - Generated Personal Access Token for ADF → Databricks authentication

5. SOURCE LAYER — Azure SQL Database

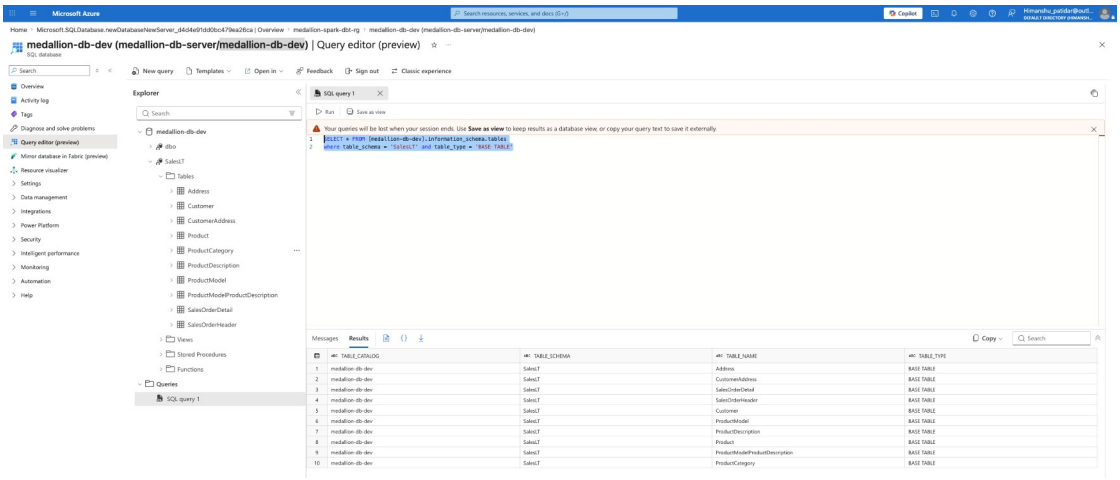

Figure 5: Azure SQL Database — SalesLT schema tables discovered via information_schema query

The project uses **AdventureWorksLT** sample database hosted on Azure SQL as the transactional source. Key tables include: Address, Customer, CustomerAddress, Product, ProductCategory, ProductDescription, ProductModel, ProductModelProductDescription, SalesOrderDetail, SalesOrderHeader.

6. ORCHESTRATION LAYER — Azure Data Factory Pipeline

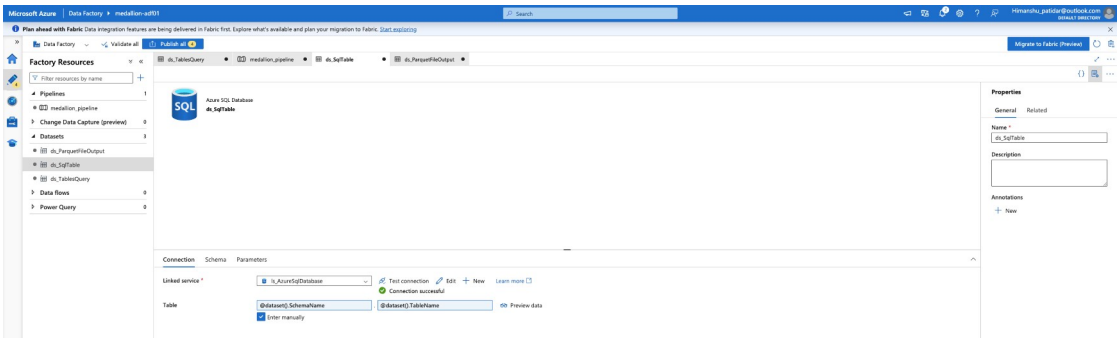

Figure 6: Azure Data Factory Pipeline datasets and linked services configuration

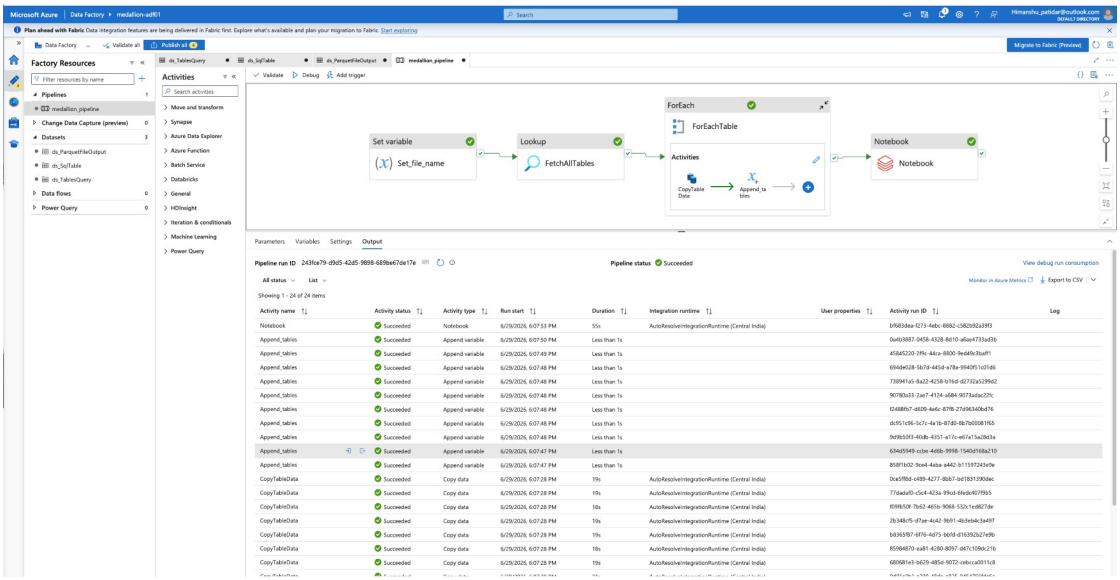

Figure 7: Azure Data Factory Pipeline 'medallion_pipeline' with Lookup + ForEach + Notebook + Copy activities, Successful pipeline execution showing all 24 activities completed with status Succeeded

Purpose: Orchestrate the entire data flow in a repeatable, scheduled, and monitorable manner. The ADF pipeline is the central orchestrator that triggers Bronze layer creation first.

- **Pipeline Activities (Correct Sequence):**
    - Lookup activity to fetch list of tables from source
    - ForEach + Append Variable to build dynamic JSON array of tables
    - Copy Data activities to load data into Bronze container in ADLS Gen2 (First output)
    - Databricks Notebook activity (at the end) to create external tables in Unity Catalog from Bronze files in container in ADLS Gen2

7. BRONZE LAYER — Raw Data Ingestion via ADF

Purpose: Store data in its original/raw form with minimal transformation. This layer acts as the system of record.

- **Implementation Flow (Triggered by ADF Pipeline):**
    - Azure Data Factory Copy Activity extracts data from Azure SQL Database and writes it to Bronze container in ADLS Gen2 as Parquet files (First output of the pipeline).
    - In the same ADF pipeline, a Notebook Activity is triggered at the end.
    - The Notebook reads the list of tables dynamically and creates external tables in Unity Catalog under the `saleslt` schema pointing to the Bronze Parquet files.

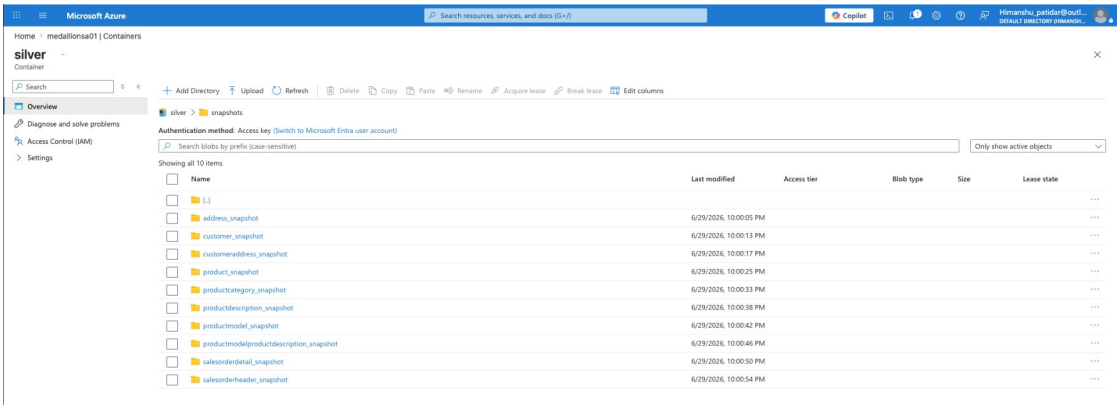

Figure 8: Bronze container in ADLS Gen2 with raw Parquet files from source tables (First output of ADF Pipeline)

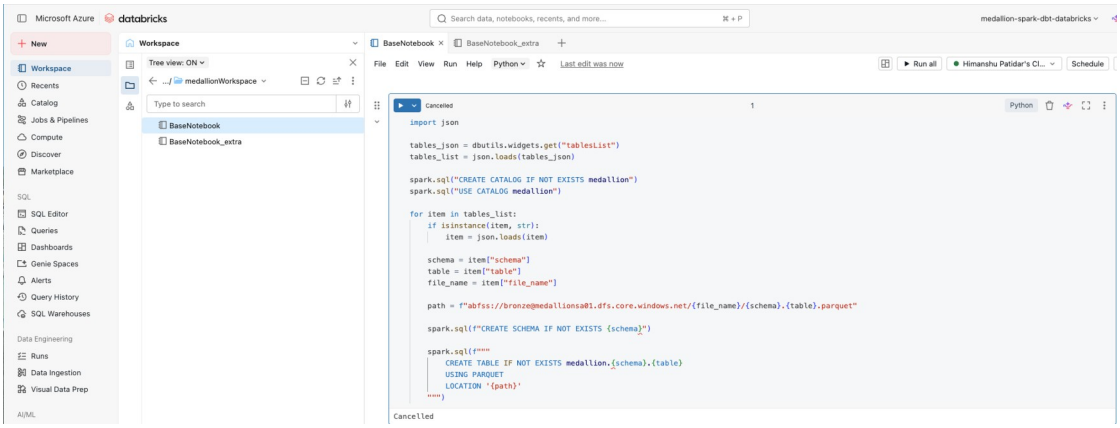

Figure 9: Databricks Notebook (BaseNotebook) that dynamically creates schemas and external tables in Unity Catalog from Bronze Parquet files (Executed after Bronze storage)

8. SILVER LAYER — dbt Snapshots (SCD Type 2)

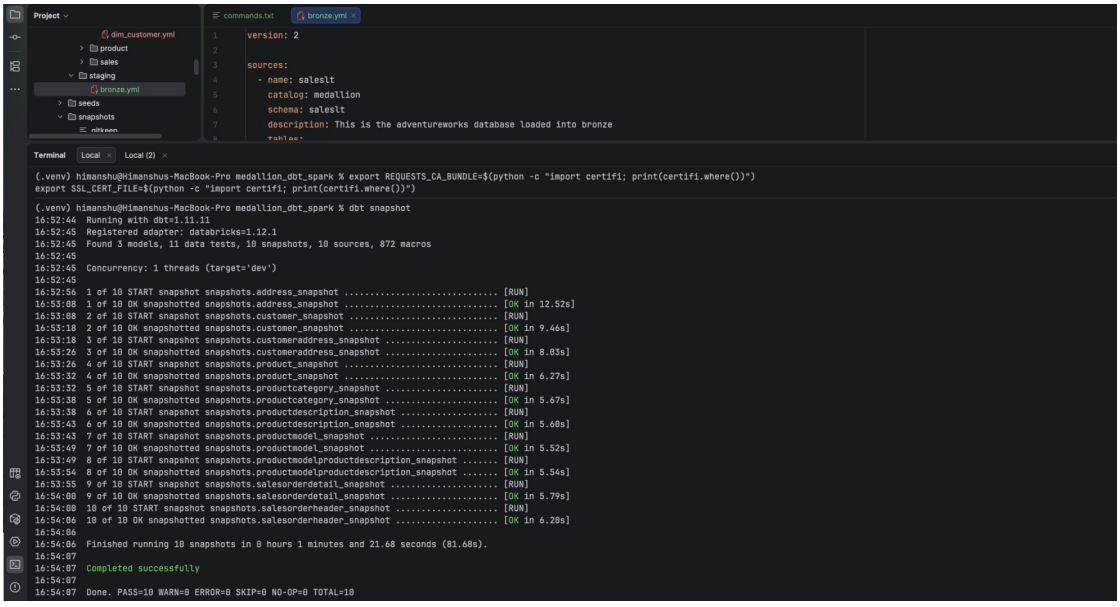

Figure 10: Successful dbt snapshot run — 10 snapshots executed (address, customer, product, salesorderdetail, etc.) in 81.68 seconds

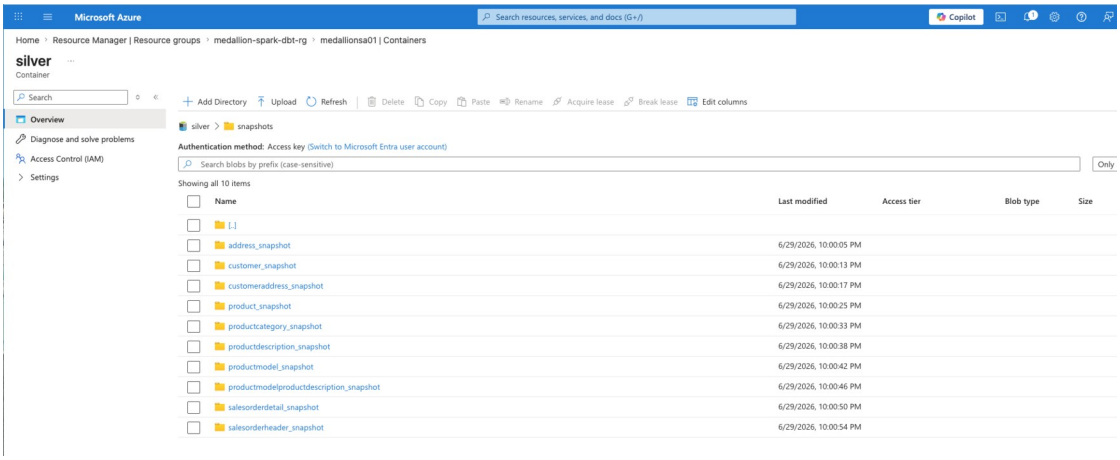

Figure 11: Silver container in ADLS Gen2 containing 10 snapshot folders with historical versions

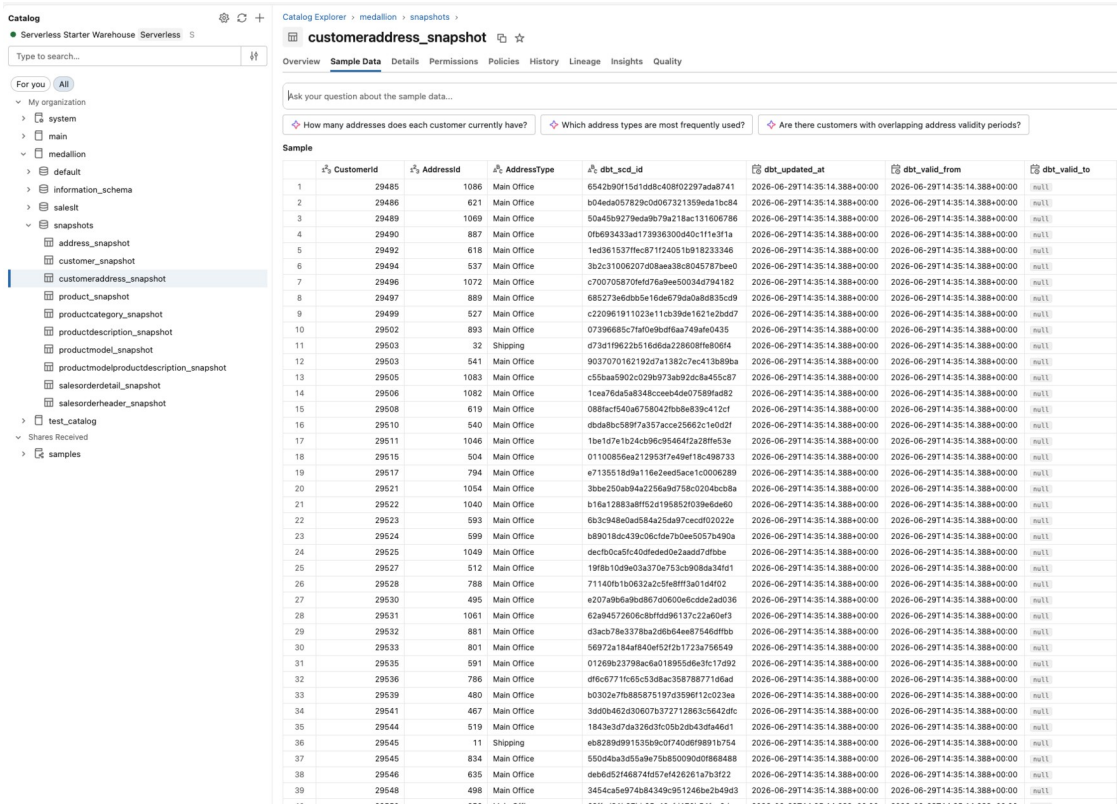

Figure 12: customeraddress_snapshot table in Unity Catalog showing SCD columns (dbt_scd_id, dbt_valid_from, dbt_valid_to)

Purpose: Capture historical changes using **Slowly Changing Dimension (SCD) Type 2** logic. This enables point-in-time analysis and maintains data lineage over time.

- **Implementation**:**
  - dbt project initialized and connected to Azure Databricks (using Databricks adapter).
  - SSL certificate issues resolved by exporting REQUESTS_CA_BUNDLE and SSL_CERT_FILE.

- 10 dbt snapshots configured against Bronze tables in Unity Catalog.
- Snapshots create versioned records in the `snapshots` schema with valid_from/valid_to timestamps.
- Data is also written to Silver container in ADLS Gen2 for physical storage.

9. GOLD LAYER — Dimensional Modeling with dbt

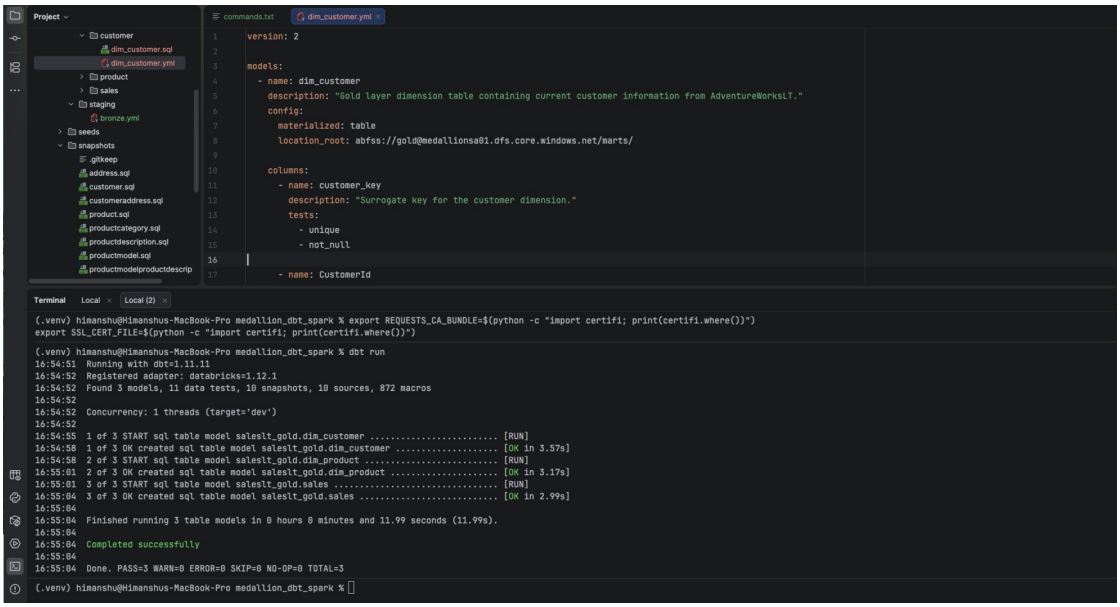

Figure 13: dbt model for dim_customer (Gold layer dimension with surrogate key and data quality tests)

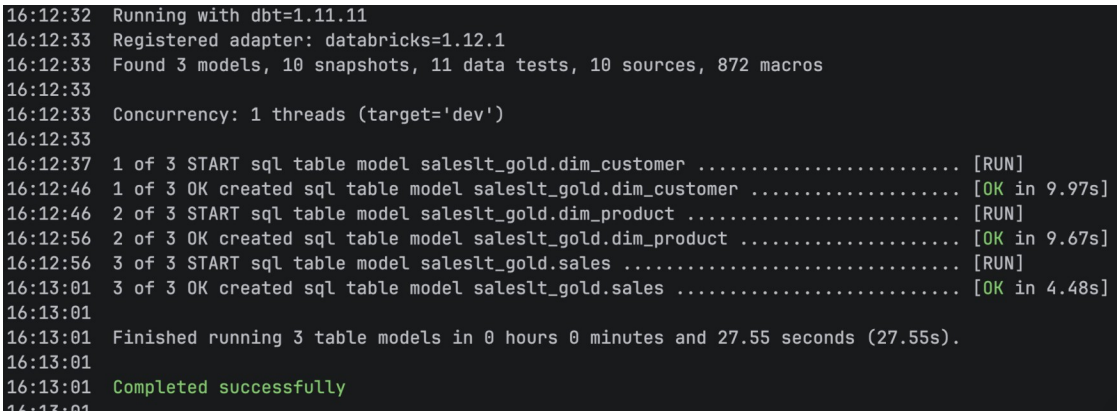

Figure 14: Successful dbt run creating 3 Gold models (dim_customer, dim_product, sales) in 11.99 seconds

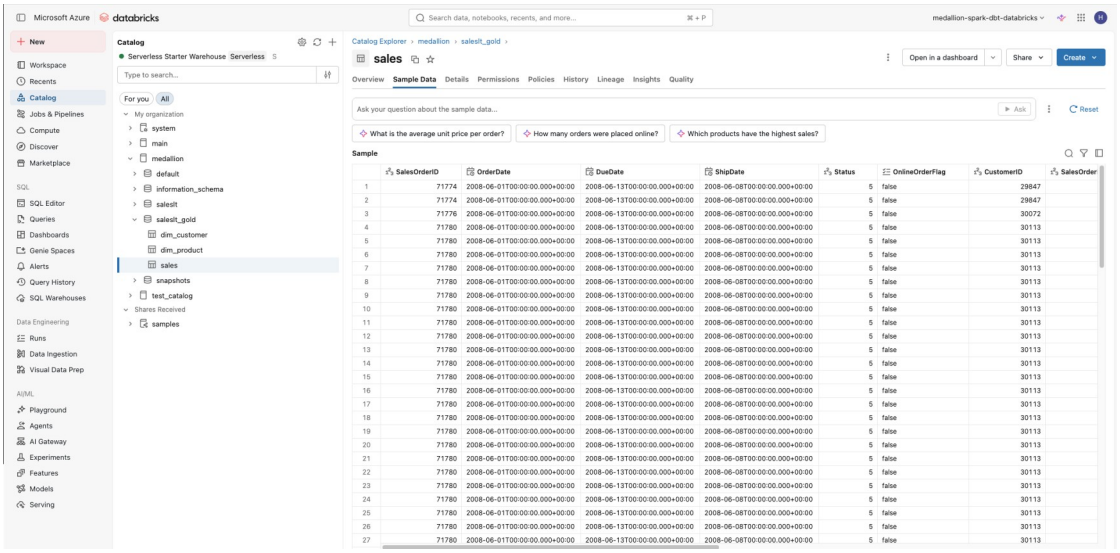

Figure 15: Gold layer 'sales' fact table in Unity Catalog under saleslt_gold schema

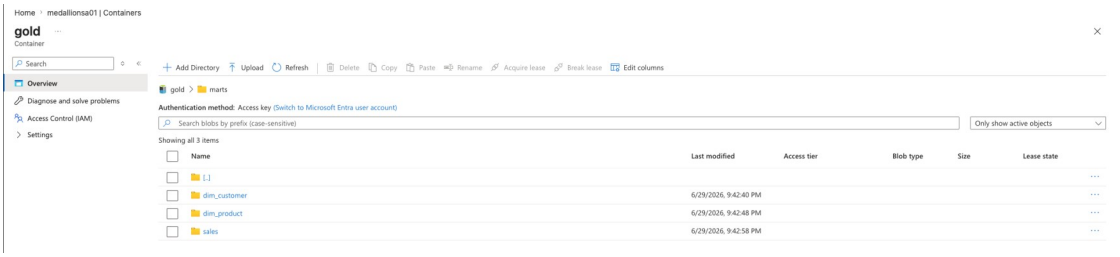

Figure 16: Gold container in ADLS Gen2 with marts/sales folder containing Delta files

Purpose: Transform cleaned and historical data into business-ready dimensional models optimized for analytics and BI.

- ****Implementation**:**
  - dbt models created in `marts` folder reading from Silver snapshots.
  - Dimension tables: dim_customer, dim_product (with surrogate keys and SCD attributes).
  - Fact table: sales (fact table with foreign keys to dimensions).
  - All models materialized as tables in the `saleslt_gold` schema in Unity Catalog.
  - Data also written to Gold container in ADLS Gen2 under marts/ folder.

10. DBT DOCUMENTATION & LINEAGE

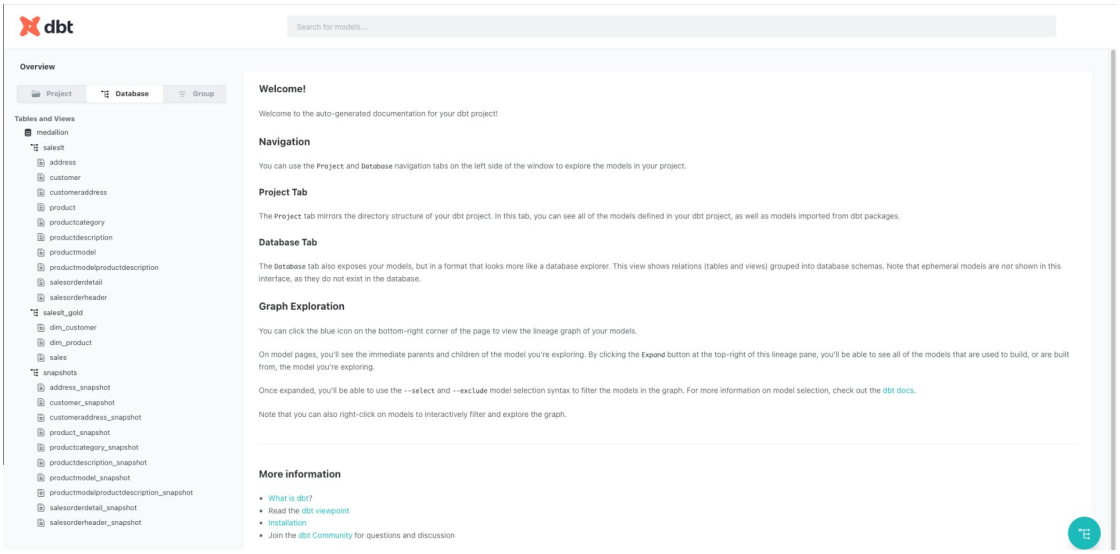

Figure 17: dbt Documentation site generated for the project

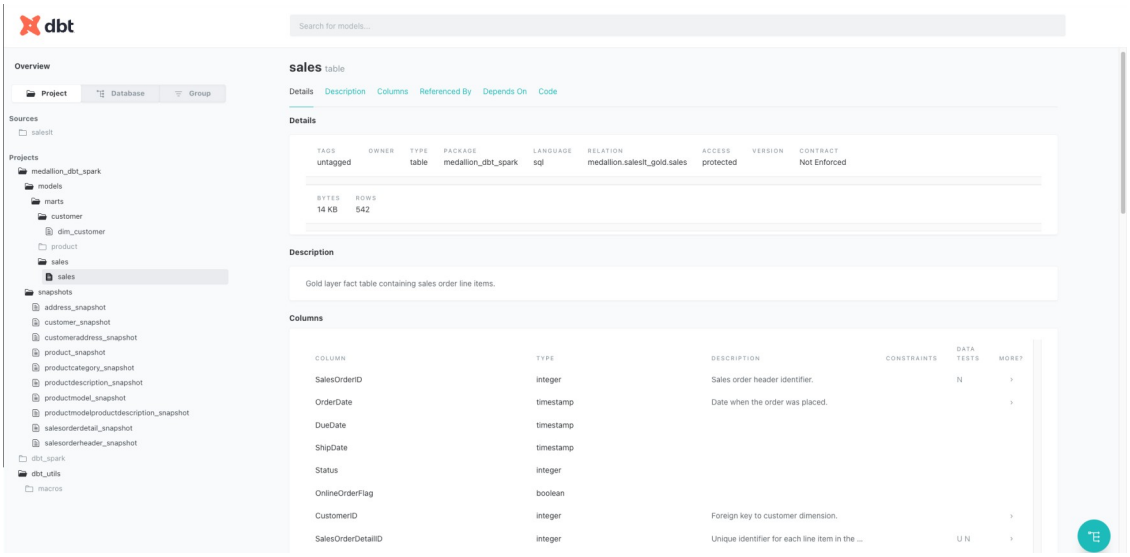

Figure 18: dbt Documentation showing models, snapshots, and lineage graph

After completing Silver and Gold layers, dbt documentation was generated to provide:

- Auto-generated documentation for all models and snapshots
- Lineage graph showing dependencies between models
- Column-level descriptions and data quality test results
- Interactive exploration of the data transformation flow

11. END-TO-END DATA FLOW

- **Step 1: Pre-requisites (Unity Catalog Setup)**
    - Create Metastore, Access Connector, Managed Identity, Storage Credential, and Catalog with schemas.

- **Step 2: Source Data Discovery**
  - Query Azure SQL Database to identify tables in SalesLT schema.
- **Step 3: Orchestration via ADF Pipeline**
  - ADF Pipeline is triggered and orchestrates the entire flow.
- **Step 4: Bronze Layer (ADF Pipeline Output)**
  - ADF Pipeline runs Copy Activity → Data loaded to Bronze container in ADLS Gen2 as Parquet (First output).
  - Notebook Activity (triggered at end of same pipeline) creates external tables in Unity Catalog under saleslt schema from Bronze files.
- **Step 5: Silver Layer (dbt Snapshots)**
  - dbt initialized and connected to Azure Databricks.
  - dbt snapshot command executed → 10 snapshots created in snapshots schema (SCD Type 2).
  - Snapshot data also stored in Silver container in ADLS Gen2.
- **Step 6: Gold Layer (dbt Models)**
  - dbt run command executed on marts → 3 models created (dim_customer, dim_product, sales).
  - Gold tables created in saleslt_gold schema in Unity Catalog.
  - Data also written to Gold container in ADLS Gen2 under marts/.
- **Step 7: Documentation & Verification**
  - dbt docs generate executed to create documentation site with lineage.
  - Final verification performed in Unity Catalog Explorer (all schemas visible).

12. KEY IMPLEMENTATION CHALLENGES & SOLUTIONS

- **Challenge 1: Unity Catalog Access Setup**
  - **Solution:** Proper sequence — Metastore → Managed Identity → Access Connector → Storage Credential with correct RBAC (Storage Blob Data Contributor).
- **:Challenge 2 dbt + Databricks SSL Issues on macOS**
  - **Solution:** Export CA bundle and SSL certificate using certifi before running dbt:
  - export REQUESTS_CA_BUNDLE=$(python -c "import certifi; print(certifi.where())")
  - export SSL_CERT_FILE=$(python -c "import certifi; print(certifi.where())")
- **Challenge 3: Dynamic Multi-Table Processing in ADF**
  - **Solution:** Use Lookup + ForEach + Append Variable (JSON array) → Pass to Databricks Notebook for dynamic external table creation.
- **Challenge 4: Snapshot vs Model Strategy**

- **Solution:** Snapshots only for SCD Type 2 dimensions. Regular models for facts and current-state dimensions.

13. KEY ACHIEVEMENTS

- Full Medallion Architecture implemented end-to-end on Azure with correct pipeline flow
- Unity Catalog properly configured with Metastore, Credentials, Access Connector, and Managed Identity
- dbt snapshots successfully implementing SCD Type 2 (10 snapshots executed)
- Clean Gold layer dimensional models with surrogate keys and data quality tests
- Robust ADF orchestration with dynamic Notebook integration for Bronze layer
- Complete governance, documentation, and lineage using Unity Catalog + dbt docs

14. SKILLS DEMONSTRATED

- Medallion / Lakehouse Architecture design and implementation
- Unity Catalog governance, metastore setup, and access management
- dbt for snapshots (SCD Type 2), modeling, testing, and documentation
- Azure Databricks + Spark for large-scale data processing
- Azure Data Factory for complex orchestration and dynamic pipelines
- Infrastructure as Code mindset for Unity Catalog and identity setup
- End-to-end data platform development from source to analytics consumption
- Troubleshooting SSL, authentication, and dynamic pipeline issues in hybrid environments

15. FINAL VERIFICATION — Unity Catalog Explorer

After completing the entire pipeline (Bronze → Silver → Gold), the final state of Unity Catalog was verified:

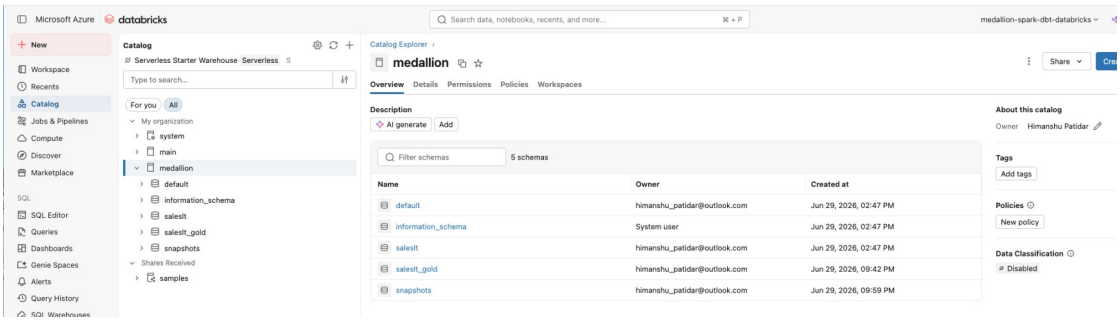

Figure 19: Unity Catalog 'medallion' with all schemas (saleslt, saleslt_gold, snapshots) — Final output after running all pipeline steps

This confirms that all layers were successfully created and governed under Unity Catalog.